Words or Vision: Do Vision-Language Models Have Blind Faith in Text?

Ailin Deng[1] Tri Cao[1] Zhirui Chen[1] Bryan Hooi[1]
[1]National University of Singapore
{ailin.deng, tricao, zhiruichen, bhooi}@nus.edu.sg

Abstract

Vision-language models (VLMs) are increasingly deployed in real-world pipelines—such as retrieval-augmented generation and autonomous agents—where extra textual context accompanies visual inputs. This context can be informative, but can also be noisy, misleading, or irrelevant. We introduce the ***Blind Faith in Text (BFiT)*** *benchmark to systematically probe how VLMs arbitrate between image and text when the two modalities conflict. BFiT augments four standard vision tasks (VQAv2, DocVQA, MathVista, and Brand Detection) with three controlled text conditions—*matched*,* corrupted*, and* irrelevant*—yielding 16,500 test samples from 5,500 base questions evaluated across ten VLMs. We introduce the* ***Text Preference Ratio (TPR)*** *to directly quantify the degree to which a model defers to text over visual evidence. Our experiments reveal a striking "blind faith" phenomenon: open-source models exhibit severe text bias under corruption (e.g., LLaVA-NeXT-7B TPR 85.52% on VQAv2), while leading proprietary models are substantially more robust (Claude Sonnet TPR 9.58%). We also investigate the factors driving this bias—instruction prompting, language model size, token order, and text relevance—and show that supervised fine-tuning with text-augmented data provides the most effective mitigation, dramatically reducing corruption accuracy losses and generalizing across datasets (LLaVA-NeXT-7B VQAv2 corruption accuracy: 28.69%→71.25%; Qwen2-VL-7B: 50.79%→82.72%). We further provide a theoretical account linking bias magnitude to the imbalance between text-only and multimodal data during training. Code and data are available at* https://github.com/d-ailin/words-or-vision.

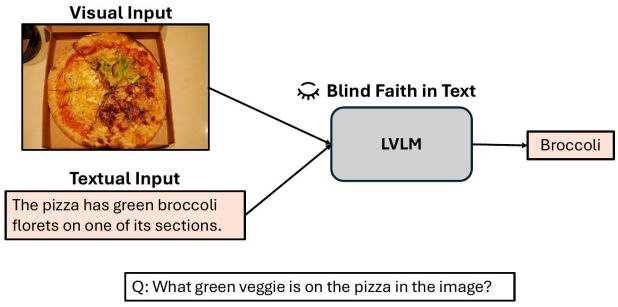

Figure 1. Illustration of the "Blind Faith in Text" (BFiT) phenomenon. When text and image conflict, many VLMs follow the (incorrect) text rather than the (correct) visual evidence. BFiT probes this modality preference by injecting matched, corrupted, and irrelevant text alongside images in standard vision tasks.

1. Introduction

Vision-language models (VLMs) have achieved remarkable progress on a wide array of multimodal tasks, including visual question answering [12], document understanding [27], and mathematical reasoning in visual contexts [26]. As VLMs become the backbone of real-world applications—such as retrieval-augmented generation [17, 40, 41], agentic systems [9, 15], and web navigation [46, 47]—they are routinely supplied with supplementary textual context alongside images. This context may be retrieved from external sources or provided by upstream pipeline stages, and it is far from guaranteed to be accurate or relevant.

A key question thus arises: *when the text and image disagree, which modality does the model trust?* Ideally, a robust VLM should recognize when accompanying text conflicts with visual evidence and rely primarily on what it "sees." However, as we show in this paper, many VLMs exhibit a systematic **blind faith in text**: they default to the information conveyed by the textual context, even when it directly contradicts the image and leads to incorrect answers.

This issue is distinct from—and complementary to—the well-studied problem of visual hallucination [14, 31, 44], where models fabricate visual content unsupported by the image. BFiT instead targets *modality arbitration*: given explicit, possibly conflicting signals from two modalities, how does the model weigh each?

Existing benchmarks predominantly evaluate VLMs on tasks where textual and visual signals are aligned [12, 18, 25, 26]. Even when adversarial or misleading text is studied, it is typically in the unimodal LLM setting [32, 39, 42].

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The multimodal modality conflict scenario has received comparatively little systematic attention.

We address this gap by introducing the **BFiT benchmark**, which augments four diverse vision tasks with three controlled text conditions and measures model responses quantitatively. Our contributions are as follows:

- We introduce BFiT, a 16,500-sample benchmark covering VQAv2, DocVQA, MathVista, and Brand Detection, each augmented with matched, corrupted, and irrelevant text (Section 2).
- We propose the **Text Preference Ratio (TPR)** metric, which directly quantifies the proportion of conflicting-text responses in which a model sides with the text over the image (Section 2.1).
- We reveal strong text bias across open-source VLMs and characterize the factors that exacerbate it: instruction style, token order, language model scale, and text relevance (Section 3).
- We demonstrate that supervised fine-tuning with text-augmented data is the most effective mitigation strategy, achieving large gains in corruption accuracy and generalizing across tasks (Section 4).
- We provide a theoretical analysis linking the observed bias to the imbalance between pure-text and multimodal training data (Section 5).

2. The BFiT Benchmark

2.1. Problem Setup and Metrics

We consider the standard multimodal question-answering setup where a model f receives an image I, optional textual context T, and a question Q, and produces an answer $\hat{a} = f(I, T, Q)$. The correct answer a^* is derivable from the image.

Text conditions. For each base instance (I, Q, a^*), we construct three variants:

- **Match**: The textual context T_{match} is a short description that, on its own, enables the correct answer a^*.
- **Corruption**: The textual context T_{corr} is a plausible but misleading description that, on its own, leads to an incorrect answer $a' \neq a^*$.
- **Irrelevance**: The textual context T_{irr} is an unrelated Wikipedia passage [28], topically disconnected from the image and question.

Metrics. We measure:

- **Base Accuracy**: performance with no textual context.
- **Accuracy**: correct answer rate under each text condition.
- **Macro Accuracy**: average accuracy across all three text conditions.
- **Normalized Accuracy (Norm)**: corruption accuracy divided by base accuracy, measuring relative performance drop.
- **Text Preference Ratio (TPR)**: fraction of corruption cases where the model's answer matches the corrupted text's wrong answer a' rather than the correct answer a^*:

$$\text{TPR} = \frac{\#\{\hat{a} = a' \wedge \hat{a} \neq a^*\}}{\#\{\hat{a} = a' \wedge \hat{a} \neq a^*\} + \#\{\hat{a} = a^*\}}.$$

A high TPR indicates that, when the model errs, it predominantly does so by following the corrupted text—a direct signal of text bias.

2.2. Data Construction

Base datasets. BFiT builds on four established benchmarks:

- **VQAv2** [12]: 1,000 samples from the validation split, covering general visual understanding.
- **DocVQA** [27]: 1,000 samples from the validation split, focusing on document images such as charts and tables.
- **MathVista** [26]: 1,000 samples from the minitest split, requiring mathematical reasoning from visual inputs.
- **Brand Detection** (KnowPhish [20]): 2,500 samples from the test split, where the model must identify a brand from a webpage screenshot and associated HTML.

These four tasks span a broad range of visual understanding difficulty and modality characteristics, yielding 5,500 base questions and 16,500 total samples (three conditions each).

Generating matched and corrupted text. For each base instance, we prompt GPT-4o [2] to produce two short textual descriptions: (1) a *matched* description that enables the correct answer without reference to the image; and (2) a *corrupted* description that is plausible and topically relevant, but leads to a different (wrong) answer. Human review confirmed that corrupted texts are coherent and not obviously incorrect on their face—they require visual grounding to refute.

Generating irrelevant text. Irrelevant passages are sampled from WikiText [28], drawn uniformly at random and filtered to ensure they share no named entities with the image content.

Prompt format. Each input follows the template:

```
[image] [cautionary instruction] [textual
context] [question]
```

where the cautionary instruction is optionally prepended to examine its effect (see Section 4).

Sanity checks. We validate the quality of generated texts by evaluating selected models on text-only inputs (no image). As shown in Table 1, matched text yields high accuracy (e.g., GPT-4o: 90.07%, Claude Sonnet: 84.39%), while corrupted text dramatically reduces accuracy (GPT-4o: 17.59%, Claude Sonnet: 16.17%) and irrelevant text also drops accuracy markedly (GPT-4o: 18.67%), confirming that each text condition behaves as intended.

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Model	VQAv2			
	Base	Match	Corruption	Irrelevance
Claude Sonnet	66.88	84.39	16.17	24.39
GPT-4o	78.39	90.07	17.59	18.67
Molmo-7B-D	76.33	88.98	18.74	35.40

Table 1. Text-only accuracy (%) across different models. It provides a sanity check for the constructed text when matched, corrupted, or irrelevant.

2.3. Evaluated Models

We evaluate ten VLMs with deterministic decoding (temperature 0):

- **Proprietary**: GPT-4o [2], GPT-4o mini, Claude Sonnet [3], Claude Haiku.
- **Open-source**: LLaVA-NeXT-7B, LLaVA-NeXT-13B, LLaVA-NeXT-34B [23], Phi3.5 [1], Molmo-7B-D [7], Qwen2-VL-7B [37].

This selection spans a range of model families, scales, and open vs. closed development paradigms.

3. Empirical Analysis

3.1. Main Results: Text Bias Under Corruption

Table 2 presents full results on VQAv2, DocVQA, and MathVista, while Table 3 covers Brand Detection. Figure 3 shows TPR across all models and conditions on VQAv2.

Proprietary models are substantially more robust. Claude Sonnet achieves the lowest TPR on VQAv2 (9.58%) and DocVQA (3.21%), and even improves slightly over its base accuracy in some corruption settings (Normalized Accuracy 101.93% on VQAv2). GPT-4o likewise achieves low TPR (27.09% VQAv2, 17.96% DocVQA) and high normalized accuracy (90.25% VQAv2, 86.59% DocVQA). These models appear to have strong visual grounding that allows them to discount misleading text.

Open-source models show severe text bias. LLaVA-NeXT-7B reaches a TPR of 85.52% on VQAv2: when the model errs under corruption, it almost always does so by following the corrupted text. This corresponds to a corruption accuracy of only 28.69%, against a base of 79.45%. LLaVA-NeXT-13B and -34B partially improve (TPR 74.43% and 67.56% respectively), but remain far behind proprietary models, suggesting that scale alone does not resolve the bias.

Large accuracy drops from corrupted text. Qwen2-VL-7B, despite strong base accuracy (85.51% VQAv2, 90.50% DocVQA), suffers substantially under corruption (normalized accuracy 59%, 64%, and 52% on VQAv2, DocVQA, and MathVista respectively). Even Phi3.5 and Molmo-7B-D, which show intermediate text-bias levels, experience significant degradation.

Real-world risk in security-critical tasks. Table 3 shows the Brand Detection results. GPT-4o and Claude Sonnet maintain near-perfect performance (TPR 0.83% and 0.96%, normalized accuracy 101.22% and 100.04%). In stark contrast, Molmo-7B-D drops to a normalized accuracy of 47.39% with a TPR of 60.40%, posing a direct security risk in phishing-detection pipelines where HTML content is deliberately crafted to mislead.

MathVista and irrelevant text. On MathVista, which requires multi-step mathematical reasoning from visual inputs, corrupted text has a particularly damaging effect: misleading numerical or symbolic information can derail the reasoning chain, compounding errors across reasoning steps. This results in steep accuracy drops even for moderately robust models (e.g., Qwen2-VL-7B normalized accuracy 52% on MathVista vs. 59% on VQAv2 under corruption). The performance gap between MathVista and other tasks suggests that reasoning-heavy benchmarks face a disproportionate risk when deployed in noisy retrieval pipelines. Under the irrelevant text condition, most models maintain near-base accuracy, suggesting effective filtering of off-topic content. Nevertheless, open-source models still exhibit mild degradation (e.g., LLaVA-NeXT-7B normalized accuracy ≈88% on VQAv2 irrelevant condition), indicating that even topically disconnected text can subtly interfere with visual grounding.

3.2. Behavior Analysis

Figure 2 categorizes each model's responses into image-consistent, text-consistent, or other, across all three text conditions. Under matched text, all models benefit from the consistent signals. Under corruption, open-source models show a dramatic shift toward text-consistent responses, while proprietary models largely maintain image-consistent behavior. Under irrelevant text, most models effectively ignore the passage and rely on the image, with some exceptions (notably Molmo-7B-D, which retains modest irrelevance sensitivity).

3.3. Factors Influencing Text Bias

We investigate four factors: (1) instruction prompting, (2) language model size, (3) token order, and (4) text relevance.

Instruction prompting. We prepend a cautionary instruction to the input, explicitly telling the model that the provided text may be unreliable. As shown in Figure 4, this marginally reduces TPR for Qwen2-VL-7B but does not eliminate the bias, indicating that simple prompting is insufficient as a mitigation strategy.

Language model size. Figure 5 shows TPR as a function of model size using LLaVA-NeXT (7B, 13B, 34B). Scaling from 7B to 34B reduces TPR (85.52%→67.56% on VQAv2 corruption), but the bias remains high even at 34B, suggesting that language model scale alone is insufficient to overcome the text preference.

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Model	VQAv2				DocVQA				MathVista			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	69.82	51.55	73.83	52.42	69.40	38.20	55.04	52.07	52.30	23.90	45.70	80.28
Claude Haiku	50.08	25.54	50.99	82.70	68.80	40.20	58.43	47.67	41.00	19.80	48.29	77.42
GPT-4o	78.39	**70.75**	90.25	27.09	85.00	73.60	86.59	17.96	**58.90**	41.20	69.95	48.98
Claude Sonnet	66.88	68.17	**101.93**	**9.58**	87.00	**84.60**	**97.24**	**3.21**	56.30	**49.30**	**87.57**	**29.14**
LLaVA-NeXT-7B	79.45	28.69	36.10	85.52	53.60	10.00	18.60	87.77	35.80	19.70	54.97	84.19
LLaVA-NeXT-13B	81.02	37.61	46.40	74.43	57.70	11.00	19.10	86.84	36.20	20.60	56.89	80.83
LLaVA-NeXT-34B	82.96	42.87	51.70	67.56	64.00	15.10	23.61	82.69	34.00	21.70	61.98	67.64
Phi3.5	75.65	35.23	46.50	74.05	78.20	50.50	64.60	40.51	43.10	22.20	51.47	80.20
Molmo-7B-D	76.33	49.29	64.50	59.40	74.00	38.40	51.90	57.20	44.90	32.90	73.27	60.63
Qwen2-VL-7B	**85.51**	50.79	59.41	29.22	**90.50**	57.50	63.63	37.41	55.40	28.90	52.18	70.23

Table 2. Performance (%) reported as Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm) and Text Preference Ratio (TPR) under corruption. **Bold**: best performance; underline: second best. Full results under all text variations are in the Appendix.

Model	Brand Detection			
	Base ↑	Corruption ↑	Norm ↑	TPR ↓
GPT-4o mini	88.84	84.8	95.44	7.48
Claude Haiku	84.40	78.72	93.27	6.44
GPT-4o	88.68	**89.76**	101.22	**0.83**
Claude Sonnet	**90.20**	90.24	100.04	0.96
LLaVA-NeXT-7B	78.60	55.32	70.39	59.17
LLaVA-NeXT-13B	83.00	60.00	72.29	40.65
LLaVA-NeXT-34B	66.28	53.52	80.77	23.49
Phi3.5	84.40	60.68	71.90	50.45
Molmo-7B-D	87.44	41.44	47.39	60.40
Qwen2-VL-7B	89.68	86.48	96.43	2.99

Table 3. Performance on the Brand Detection dataset reported in Base Accuracy, Corruption Accuracy, Normalized Corruption Accuracy (Norm), and Text Preference Ratio (TPR). **Bold**: best performance; underline: second best performance.

Token order. We test whether placing the text tokens *before* vs. *after* the image affects text bias, using Phi3.5. As shown in Figure 6, placing text before the image significantly increases TPR, consistent with findings that language models are sensitive to the position of information in context [24].

Text relevance. We distinguish two types of "corrupted" text: *topically relevant* (consistent topic but wrong answer) versus *topically irrelevant* (unrelated topic). As shown in Figure 7, seemingly relevant text—even when incorrect—induces stronger text preference than clearly irrelevant text. This mirrors findings in LLM knowledge conflict studies [39, 42] and highlights that relevance cues amplify credulity.

Uni-modal certainty. Figures 8a and 8b show that the model's modality preference under conflict is correlated with its confidence in each modality individually. When GPT-4o is highly confident in the visual answer (high image-only certainty), it tends to remain image-consistent even under corruption. By contrast, LLaVA-NeXT-7B exhibits text-consistent behavior regardless of visual certainty, reflecting its strong prior toward text.

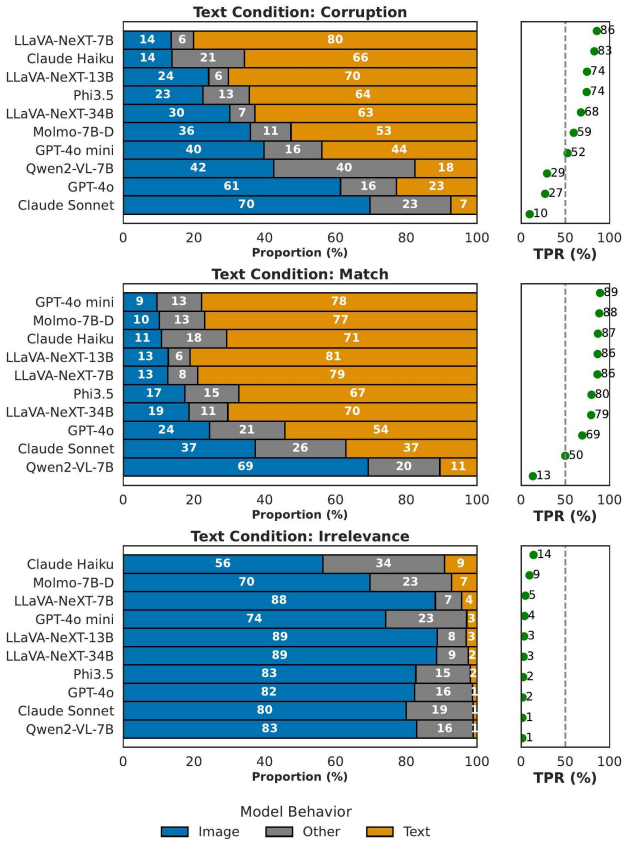

Figure 2. Model prediction behaviors—image-consistent, text-consistent, or other—across all three text conditions and ten models. Under corruption, open-source models shift heavily toward text-consistent behavior.

4. Investigated Solutions

We investigate two mitigation strategies: (1) instruction-based prompting and (2) supervised fine-tuning (SFT) with text-augmented data.

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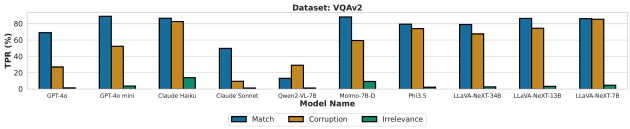


Figure 3. Text Preference Ratio (TPR) of all models under matched, corrupted, and irrelevant text on VQAv2. Corrupted text induces the strongest text bias, especially in open-source models.

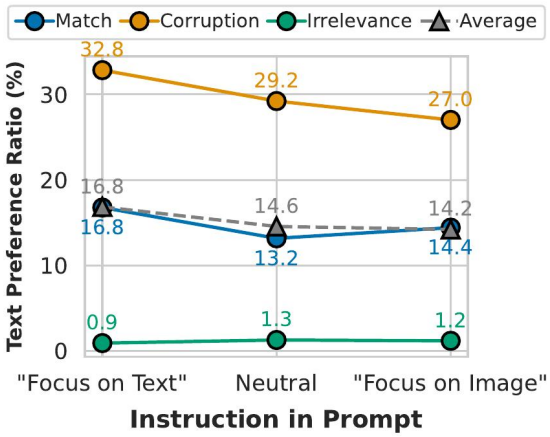


Figure 4. Effect of cautionary instruction prompting on TPR for Qwen2-VL-7B across text conditions. Prompting slightly reduces corruption bias but cannot fully address it.

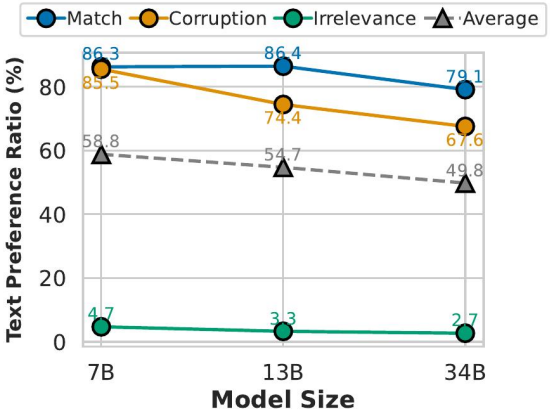


Figure 5. Effect of language model size on TPR using LLaVA-NeXT (7B, 13B, 34B). Larger models show reduced but still substantial text bias.

4.1. Instruction Prompting

We prepend a cautionary instruction to the input:

Cautionary Instruction

"Please note that the provided text information may not be accurate. Use it only as reference and prioritize information from the image when answering."

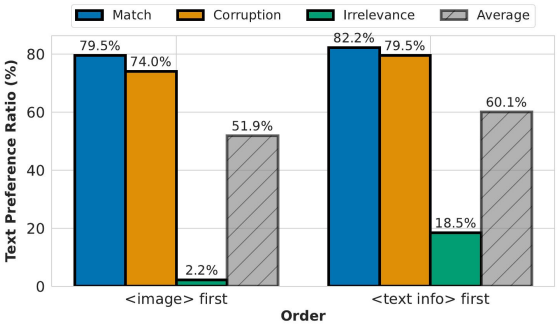


Figure 6. Effect of token order on text bias in `Phi3.5`. Placing text tokens before image tokens increases TPR under corruption.

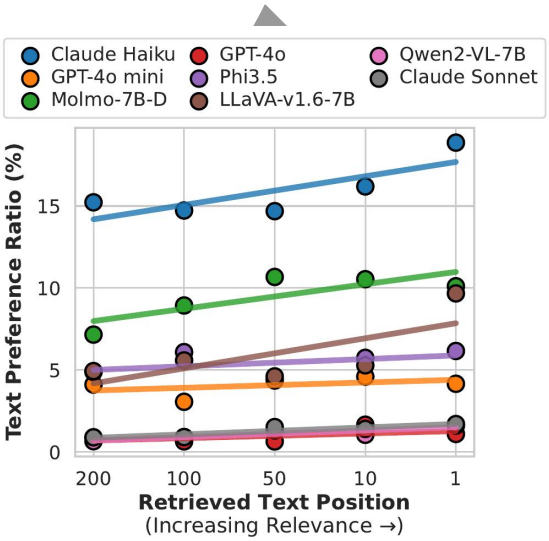
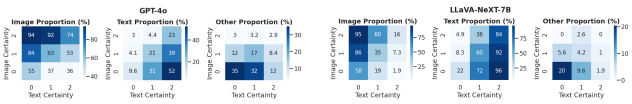


Figure 7. Effect of text relevance on TPR. Topically relevant (but wrong) text increases text bias more than irrelevant text.



(a) GPT-4o (b) LLaVA-NeXT-7B

Figure 8. Effect of uni-modal certainty on modality preference. GPT-4o relies on vision when visually certain; LLaVA-NeXT-7B defaults to text regardless.

As reported in Table 4, instruction prompting yields modest improvements in VQAv2 corruption accuracy (LLaVA-NeXT-7B: 28.69%→34.27%; Qwen2-VL-7B: 50.79%→54.78%) without affecting base performance. While statistically meaningful, these gains are far below what is needed for reliable deployment.

4.2. Supervised Fine-Tuning

We construct a fine-tuning dataset by augmenting VQAv2 training instances with all three text conditions (matched,

Model	VQAv2				
	Base ↑	Match ↑	Corruption ↑	Irrelevance ↑	Macro ↑
LLaVA-NeXT-7B	**79.45**	**92.32**	28.69	**79.43**	66.81
Instruction	**79.45**	92.25	34.27	78.15	68.22
SFT	77.48	87.56	**71.25**	77.32	**78.71**
Qwen2-VL-7B	**85.51**	**92.76**	50.79	83.70	75.75
Instruction	**85.51**	92.62	54.78	82.82	76.74
SFT	84.18	87.01	**82.72**	**84.00**	**84.58**

Table 4. In-distribution performance comparison between original models, instruction and fine-tuned models.

Model	DocVQA		MathVista		Brand Detection	
	Base ↑	Macro ↑	Base ↑	Macro ↑	Base ↑	Macro ↑
LLaVA-NeXT-7B	**53.60**	51.07	**35.80**	41.03	78.60	46.44
Instruction	**53.60**	49.27	**35.80**	41.20	78.60	47.36
SFT	52.20	**56.17**	35.30	**41.63**	**81.36**	**72.29**
Qwen2-VL-7B	**90.50**	80.83	55.40	53.87	**89.68**	81.85
Instruction	**90.50**	80.77	55.40	54.10	**89.68**	84.48
SFT	90.30	**88.97**	**58.50**	**57.17**	89.44	**88.75**

Table 5. Performance comparison with Base and Macro accuracy based on DocVQA, MathVista, and Brand Recognition. See full results under different text conditions in Appendix.

corrupted, irrelevant), using the same GPT-4o text-generation pipeline as BFiT. Models are trained to answer correctly regardless of text condition. We apply SFT to LLaVA-NeXT-7B and Qwen2-VL-7B.

In-distribution results (VQAv2). Table 4 shows that SFT dramatically reduces text bias. LLaVA-NeXT-7B's corruption accuracy rises from 28.69% to 71.25% (Macro Accuracy: 66.81%→78.71%). Qwen2-VL-7B improves from 50.79% to 82.72% corruption accuracy (Macro: 75.75%→84.58%). Base accuracy drops only marginally (LLaVA-NeXT-7B: 79.45%→77.48%; Qwen2-VL-7B: 85.51%→84.18%), indicating a favorable trade-off.

Out-of-distribution generalization. Table 5 shows that SFT generalizes across tasks. On DocVQA, LLaVA-NeXT-7B's Macro Accuracy improves from 51.07% to 56.17%; on Brand Detection, it rises from 46.44% to 72.29%. For Qwen2-VL-7B, DocVQA Macro increases from 80.83% to 88.97% and Brand Detection from 81.85% to 88.75%. These improvements are achieved without training on data from these tasks, demonstrating the broad applicability of the learned modality-balancing behavior.

Effect of text-only data in SFT. Figure 9 investigates the role of the amount of text-augmented data used in fine-tuning. As training data increases, corruption accuracy rises sharply while base accuracy is mostly preserved, confirming that more text-augmented data is beneficial up to a saturation point.

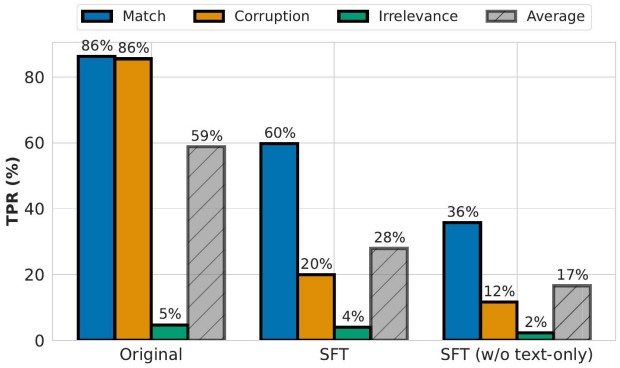


Figure 9. Effect of training data size on SFT for LLaVA-NeXT-7B. Left: TPR as a function of text-augmented training samples. More data consistently reduces text bias without sacrificing base performance.

5. Theoretical Analysis

We provide a theoretical account of why VLMs tend to exhibit text bias, grounding it in the structure of typical multimodal pretraining. Let P_{text} and P_{vis} denote the model's probability distributions conditioned on text-only and vision-only inputs respectively. During pretraining, large VLMs are exposed to a mixture of pure-text data and multimodal data. Let $\lambda \in [0, 1]$ denote the fraction of multimodal examples in training.

Theorem 5.1 (Informal)**.** *Suppose a VLM is pretrained on a mixture with fraction λ of multimodal examples and $(1-\lambda)$ of text-only examples. Under mild regularity assumptions, the model's output distribution under conflicting text and image can be decomposed as:*

$$P_{\text{model}}(a \mid I, T_{\text{corr}}, Q) \approx \lambda \cdot P_{\text{vis}}(a \mid I, Q) + (1-\lambda) \cdot P_{\text{text}}(a \mid T_{\text{corr}}, Q).$$

When $\lambda \ll 1$, the model heavily weights the text component, predicting the corrupted answer with high probability.

This result suggests that the dominance of text-only data in pretraining corpora—which is well-documented [4, 30]—naturally leads to a text-over-vision preference when the two modalities conflict. The mixture weight λ is the key parameter: models trained with a higher proportion of multimodal data (such as proprietary VLMs, which are believed to use significantly more image-text paired data) should exhibit lower TPR.

This theory also explains why SFT is effective: fine-tuning on multimodal examples with conflicting text effectively increases the effective λ for conflict scenarios, directly training the model to down-weight misleading text signals. It further explains why instruction prompting provides only marginal gains—adjusting the prompt changes the surface-level input but does not alter the underlying model parameters that encode the text preference.

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The framework additionally predicts that bias severity should vary with task characteristics. Tasks dominated by language priors (e.g., VQAv2) are more susceptible than tasks where the image provides a self-contained grounding signal (e.g., DocVQA, where document text is itself a visual element), since the model's image-only certainty is lower in the former. This is consistent with our observations: LLaVA-NeXT-7B achieves a higher TPR on VQAv2 (85.52%) than on DocVQA (69.83%). Furthermore, the theory predicts that proprietary models—believed to use substantially larger fractions of image-text paired data during pretraining—should naturally exhibit lower TPR, in line with the empirical gap between Claude Sonnet (TPR 9.58%) and open-source counterparts.

The theoretical framework can be further extended to account for the effect of instruction-following fine-tuning (RLHF). Post-training alignment procedures optimize models toward human preferences; if annotators consistently prefer text-consistent responses (e.g., because they appear more confident or are easier to verify), RLHF may inadvertently increase the effective weight on P_{text} beyond what pretraining alone would predict. This would explain why instruction-tuned open-source VLMs exhibit strong text bias even when their base models retain reasonable visual grounding, and why the performance gap between open-source and proprietary models persists despite rapid capability gains on standard benchmarks.

6. Related Work

VLM benchmarks. Numerous benchmarks evaluate VLMs on diverse visual tasks [5, 12, 18, 21, 25–27, 35]. However, these benchmarks typically present aligned image-text pairs and do not probe modality conflict scenarios. BFiT fills this gap by systematically introducing controlled textual inconsistencies.

Hallucination in VLMs. A related line of work studies how VLMs hallucinate visual content [14, 31, 36, 43, 44]. Proposed mitigations include improved decoding [6, 8, 16], contrastive methods [19], and training-time interventions [22, 45]. Our work differs in that we study the complementary failure mode of over-trusting externally provided text.

Knowledge conflicts in LLMs. Several works study how LLMs handle conflicts between parametric knowledge and retrieved context [24, 29, 32, 33, 39, 42]. Wu et al. [39] show that LLMs adopt incorrect retrieved content over 60% of the time. We extend this investigation to the multimodal setting, where the competing signal is visual rather than parametric.

Retrieval-augmented VLMs. RAG approaches [17, 34, 40, 41] augment VLMs with retrieved context to improve factuality. Our findings highlight a key risk in such pipelines: when retrieved text is inaccurate (a common occurrence in noisy retrieval), models may blindly follow it at the expense of visual evidence.

Adversarial robustness. BFiT is related to adversarial attacks on VLMs [10, 11, 13, 38], but focuses on naturalistic text corruptions rather than imperceptible perturbations. Our corrupted text is human-readable and resembles plausible context, making it more representative of realistic deployment failures.

7. Conclusion and Discussion

We introduced BFiT, a benchmark for probing modality preference in VLMs, and demonstrated a widespread "blind faith in text" phenomenon across open-source models. Key findings include: (1) open-source VLMs exhibit severe text bias under conflicting text (TPR up to 85%), while proprietary models are substantially more robust; (2) factors such as token order, text relevance, and LM scale modulate the bias; and (3) supervised fine-tuning with text-augmented data is the most effective mitigation, yielding large gains in corruption accuracy and generalizing across tasks.

Practical recommendations. Our results carry concrete design implications for practitioners. When deploying VLMs in retrieval-augmented pipelines, one should (i) prefer models fine-tuned on text-augmented data, as SFT provides the most reliable mitigation; (ii) place image tokens before text tokens in the prompt to reduce position-induced text bias; and (iii) adopt TPR as a standard evaluation metric alongside accuracy, since corruption accuracy alone does not distinguish robust visual grounding from text-driven errors. More broadly, our findings suggest that the modality composition of pretraining data is a primary driver of text bias, motivating the development of pretraining curricula that maintain a higher proportion of multimodal examples relative to text-only data. Benchmark design should also evolve: future VLM evaluations would benefit from routinely including corrupted-text conditions to stress-test modality arbitration, complementing the prevalent aligned-context paradigms.

Limitations. Our corrupted texts are generated by GPT-4o, which may not cover all types of naturalistic errors. The theoretical analysis relies on simplifying assumptions about the pretraining mixture. Future work should investigate more diverse corruption types, closed-source model internals, and multi-turn or agent-based settings. The interplay between RLHF-based alignment and modality bias also remains underexplored; post-training procedures that reward helpfulness may inadvertently reinforce text deference if annotators favor fluent text-consistent answers.

Broader impact. The blind faith phenomenon poses a concrete risk in deployed systems: malicious actors could craft misleading text to manipulate VLM responses, particularly in security-sensitive applications such as phishing detection. Our findings underscore the importance of building

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VLMs with calibrated, modality-aware trust mechanisms. They also suggest that standard evaluation protocols for VLMs should routinely include modality-conflict scenarios, since benchmarks that rely exclusively on aligned image-text pairs may substantially overestimate real-world robustness when models are deployed in retrieval-augmented or agentic pipelines.

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